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How green knowledge-oriented leadership drives green innovation in SMEs: the mediating role of environmental strategy and the moderating role of green AI capability

Ibraheem Saleh Al Koliby[a,b], Abdullah Kaid Al-Swidi[c], Mohammed A. Al-Hakimi[b,d] and Shaima Ali Galeb Farhan[e]

[a]School of International Studies, Universiti Utara Malaysia, Sintok, Kedah, Malaysia; ; [b]Marketing and Production Department, Faculty of Administrative Sciences, Thamar University, Dhamar, Yemen; [c]Department of Management and Marketing, College of Business and Economics, Qatar University, Doha, Qatar; [d]Korea University Business School, Korea University, Seoul, South Korea; [e]School of Management, University of Science and Technology of China, Anhui, China

ABSTRACT

This study examines how green knowledge-oriented leadership (GKOL) drives green innovation (GI) in manufacturing SMEs, with a focus on the mediating role of environmental strategy (ES) and the moderating effect of green artificial intelligence capability (GAIC). Drawing on the Natural Resource-Based View (NRBV) and Dynamic Capability Theory (DCT), the study developed and empirically tested an integrated framework that captures the interplay between leadership, strategy, and digital capability in promoting sustainable innovation. Data were collected from 219 Malaysian manufacturing SMEs using a structured questionnaire, and structural equation modeling was employed via SmartPLS to evaluate the proposed relationships, with the reliability and validity of the constructs verified through composite reliability, average variance extracted, and discriminant validity. The findings reveal that GKOL significantly enhances ES, which in turn enhances GI, with ES partially mediating this relationship. Additionally, GAIC strengthens the effect of GKOL on GI, underscoring the role of AI-enabled capabilities in amplifying green leadership outcomes. This study contributes to the literature by offering a unified leadership–strategy–technology framework for understanding sustainability transformation in resource-constrained SME settings and provides actionable insights for managers and policymakers on leveraging GKOL and digital transformation for sustainable development.

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1. Introduction

Amid growing concerns over climate change, resource scarcity, and increasing regulatory and stakeholder pressures, environmental sustainability has become a critical challenge for firms worldwide. In this evolving landscape, Green Innovation (GI) the development and implementation of eco-friendly products, processes, and practices has emerged as a strategic response to enhance both environmental performance and competitive advantage (Al-Hakimi et al., 2022; Mulaessa & Lin, 2021; Solovida & Latan, 2017). GI not only helps firms comply with environmental regulations but also reduces operational costs, enhances brand image, and addresses rising stakeholder expectations for sustainability (Fahad et al., 2022). For small and medium-sized enterprises (SMEs), especially in the manufacturing sector, GI presents a pathway to address ecological challenges and respond to rising stakeholder expectations (Al-Swidi et al., 2022). However, SMEs especially in developing nations, such as Malaysia often struggle to adopt GI due to limited financial resources, low environmental awareness, and lack of technical capabilities (Al Koliby et al., 2024b). Compared to large firms, SMEs face considerable barriers to sustainability transformation, highlighting the urgent need for supportive leadership and organizational capabilities that can facilitate the green innovation process.

CONTACT Mohammed A. Al-Hakimi alhakimi111@gmail.com Marketing and Production Department, Faculty of Administrative Sciences, Thamar University, Dhamar, Yemen

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Green Knowledge-Oriented Leadership (GKOL) has recently gained prominence as a leadership style capable of driving sustainable transformation within firms. GKOL a leadership approach focused on creating, sharing, and applying environmental knowledge within organizations to drive sustainability-oriented decisions (Naqshbandi & Jasimuddin, 2018). GKOL empowers employees to explore green practices, fosters a shared environmental vision, and aligns organizational values with ecological responsibility (Donate & de Pablo, 2015; Khan et al., 2023). In the context of SMEs, where leadership plays a pivotal role in setting priorities and overcoming institutional voids, GKOL is especially crucial for instigating change and innovation (Begum et al., 2022; Zhang et al., 2024). Yet, prior research presents mixed findings on the relationship between GKOL and GI while some studies report a positive effect (e.g. Idrees et al., 2023), others suggest that leadership alone may not be sufficient to generate meaningful innovation outcomes in SMEs due to structural and technological limitations (e.g. Ahmad et al., 2024). This indicates a research gap in understanding how GKOL translates into green innovation outcomes, particularly in resource-constrained SME environments.

To address this gap, the present study explores the mediating role of environmental strategy (ES) a firm's long-term strategic orientation toward environmental responsibility in channeling the influence of GKOL into tangible innovation. ES provides a structured framework that guides green initiatives by aligning leadership vision with innovation efforts, embedding sustainability into business policies, and optimizing resource allocation (Shehzad et al., 2021, 2024). By setting clear sustainability goals, compliance mechanisms, and performance metrics, ES bridges the gap between green knowledge leadership and innovation outcomes, ensuring that GI is strategically implemented (Aftab et al., 2022; Mishra & Yadav, 2021). Empirical evidence suggests that firms with proactive ES are more likely to succeed in GI, as this strategy enhances resource efficiency, pollution prevention, and sustainable product development (Aftab et al., 2023; Kraus et al., 2020). However, the mediating role of ES between GKOL and GI remains underexplored, particularly in the SME sector, where strategic planning is often informal or reactive. This study addresses this gap by investigating whether ES acts as a mechanism through which green knowledge-oriented leaders can enhance GI performance in resource-constrained SME environments.

Beyond ES, technological capabilities are also essential in reinforcing leadership-driven innovation. In this regard, the study introduces green artificial Intelligence capability (GAIC) as a moderating variable that enhances the impact of GKOL on GI. GAIC refers to an organization's ability to leverage artificial intelligence (AI) technologies such as data analytics, machine learning, and automation to process environmental data, predict sustainability trends, and automate green solutions (Abdulmuhsin et al., 2025; Wilson et al., 2023). AI-driven capabilities are particularly beneficial for SMEs, as they help overcome resource constraints and facilitate evidence based decision-making (Murtinu & De Massis, 2025; Ueasangkomsate, 2025). Particularly for Malaysian SMEs facing technological and human capital limitations, GAIC can serve as a strategic enabler that enhances the impact of leadership on green innovation. However, the moderating role of GAIC in the relationship between GKOL and GI has not been sufficiently examined (Al Halbusi et al., 2025; Salehzadeh et al., 2024), representing a critical theoretical and empirical gap. Exploring this dynamic interaction can provide deeper insights into how digital capabilities strengthen sustainability outcomes in SMEs.

To address these gaps, this study developed a comprehensive model that integrates GKOL, ES, GAIC, and GI in the context of manufacturing SMEs in Malaysia, using the Natural Resource-Based View (Hart, 1995) and Dynamic Capability Theory (Teece et al., 1997) as theoretical lenses. To sum up, the study seeks to answer three central questions:

RQ1: How does GKOL influence GI?
RQ2: Does ES mediate the relationship between GKOL and GI? and
RQ3: Does GAIC moderate the relationship between GKOL and GI?

By answering these questions, the study contributes to a more nuanced understanding of the leadership-strategy-technology nexus in driving sustainable innovation within SMEs. Theoretically, it enriches the literature on sustainability-oriented leadership and green innovation by clarifying the roles of ES and GAIC as critical mediating and moderating mechanisms. Empirically, it presents evidence from a developing country context Malaysia where SMEs are central to industrial growth and environmental

impact, providing actionable insights for SME leaders and policymakers on how to cultivate green leadership, develop strategic environmental plans, and harness AI capabilities to foster innovation and achieve long-term sustainability. The remainder of this paper is structured as follows. Section 2 presents the literature review, encompassing the theoretical underpinnings of the study and the hypotheses development. Section 3 outlines the research methodology, while Section 4 reports the empirical results. Section 5 discusses the theoretical and practical implications of the findings, offering insights for both academic researchers and practitioners. Finally, Section 6 presents conclusions, along with the limitations and avenues for future research.

2. Literature review

2.1. Theoretical basis

This study builds upon two key theoretical frameworks the Natural Resource-Based View (NRBV) and the Dynamic Capability Theory (DCT) to examine how internal leadership, strategy, and technological capabilities collectively drive GI in SMEs.

According to NRBV, organizational capabilities that are aligned with environmental sustainability such as green leadership and strategy can serve as valuable, rare, inimitable, and non-substitutable (VRIN) resources that yield long term competitive advantage (Hart, 1995). In this context, GKOL is conceptualized as a rare internal capability that promotes the generation and diffusion of environmentally focused knowledge, enabling sustainable transformation within firms (Naqshbandi & Jasimuddin, 2018). NRBV also justifies the importance of ES as a deliberate resource deployment mechanism that facilitates the execution of eco-innovation objectives through structured policies and performance metrics (Ko & Liu, 2017).

Meanwhile, DCT complements NRBV by emphasizing the firm's ability to adapt, integrate, and reconfigure resources in response to dynamic environmental and technological changes (Teece et al., 1997). GAIC aligns with DCT as it reflects the organization's dynamic capability to leverage AI tools such as predictive analytics, automation, and real-time monitoring to enhance decision-making and support GI under uncertain conditions (Salehzadeh et al., 2024). In this context, recent work by Murtinu and De Massis (2025) contributes a useful lens by describing AI machines as relational nonhuman actors in entrepreneurial teams, supporting our view of GAIC as a collaborative digital enabler in SMEs.

While prior studies have independently linked leadership styles, strategic environmental orientations, and AI capabilities to green outcomes, a significant theoretical gap remains in understanding how these elements interact (e.g. Al-Swidi et al., 2024; Khan et al., 2023; Salehzadeh et al., 2024). Specifically, the mediating role of ES and the moderating role of GAIC in the GKOL–GI relationship have not been adequately explored in SME contexts particularly in developing economies, such as Malaysia. By addressing these interconnections, this study enriches the NRBV and DCT perspectives by providing a more integrated and dynamic understanding of how leadership, strategy, and digital capabilities synergize to foster GI.

2.2. Green knowledge-oriented leadership and green innovation

Knowledge-oriented leadership (KOL) is defined as 'an attitude or action, observed or imputed, that prompts the creation, sharing, and utilization of new knowledge in a way that seems to bring a shift in thinking and collective outcomes' (Naqshbandi & Jasimuddin, 2018, p.702). Based on this, GKOL can be defined as an attitude or action, observed or imputed, that fosters the creation, sharing, and utilization of environmentally sustainable knowledge, promoting shifts in thinking and collective outcomes that align with ecological responsibility and sustainable development goals. GKOL significantly influences the development and application of green knowledge within organizations (Idrees et al., 2023). Leaders who prioritize environmental sustainability integrate eco-conscious practices into their decision-making, fostering a culture that values green knowledge (Shafait & Huang, 2024). From the perspective of the NRBV, green knowledge is a strategic intangible resource that can yield environmental and competitive advantages when effectively managed (Sahoo et al., 2023). GKOL enables firms to identify, develop, and utilize

such green knowledge, fostering eco-innovation (Shafait & Huang, 2024). By promoting education, innovation, and collaboration, GKOL encourages employees to adopt sustainable practices and enhances overall organizational commitment to sustainability (Mittal & Dhar, 2016). In this context, Abourokbah et al. (2024) assert that green leadership cultivates a workplace culture centered on learning and adaptability, empowering employees to create innovative responses to environmental issues and shifting market needs. Indeed, leaders invest in training and collaboration with external experts, empowering teams with the skills necessary for addressing environmental challenges. This not only builds expertise but also strengthens the organization's competitive edge in sustainability (Khan et al., 2023; Riva et al., 2021). Furthermore, GKOL bridges the gap between green knowledge and its practical application. In this context, Khan et al. (2023) assert the role of sustainability-focused leadership in fostering green creativity and innovation. Leaders reinforce sustainability through their actions and decision-making, ensuring that green knowledge is implemented effectively within the organization. This alignment of goals with environmental objectives is key to translating theory into practice and fostering long-term sustainability (Farhan, 2024; Tan et al., 2020). Based on the above, we propose the following:

H1. *GKOL significantly predicts GI.*

2.3. Green knowledge-oriented leadership and environmental strategy

GKOL plays a critical role in shaping an organization's ES by embedding sustainability-oriented values and knowledge into formal strategic objectives. Indeed, leaders focusing on green knowledge drive the adoption of eco-friendly practices, such as green technologies and resource efficiency, aligning the organization's strategy with environmental sustainability (Begum et al., 2022; Khan et al., 2023). Drawing from the NRBV, leadership's emphasis on ecological values mobilizes internal resources toward achieving environmental goals, positioning ES as a capability that aligns with stakeholder and regulatory expectations (Hart, 1995). This approach ensures that ES is comprehensive and responsive to emerging challenges (Abbas & Khan, 2023; Wang et al., 2022). Additionally, GKOL aligns organizational objectives with environmental goals, ensuring that resources are allocated to support green initiatives. In this regard, DCT explains how GKOL enhances a firm's capacity to realign strategies in response to changing environmental challenges. This creates actionable, effective strategies that promote both sustainability and business success (Banmairuroy et al., 2022; González-Mohíno et al., 2024; Naqshbandi & Jasimuddin, 2018).

Numerous studies suggest that green leadership significantly advances corporate ES (Hussain et al., 2022). As Su et al. (2020) highlight, green leadership significantly benefits from the acquisition and application of green knowledge, which enhances the capacity to design and implement effective green strategies and initiatives. Furthermore, knowledge-based leadership plays a crucial role in advancing environmental strategies by promoting and streamlining green knowledge management. This leadership approach leads to more sustainable strategic outcomes, aligns with the expectations of eco-conscious consumers, and supports the delivery of high-quality, cost-effective products and services through the efficient use of natural resources (Al-Faouri, 2023). On this basis, the following exploration is directed toward framing the hypotheses:

H2. *GKOL significantly predicts ES.*

2.4. Environmental strategy and green innovation

GI involves developing eco-friendly products, technologies, and practices to reduce environmental impacts and improve sustainability (Mulaessa & Lin, 2021; Soewarno et al., 2019). Researchers have increasingly focused on proactive ES, which helps firms achieve better sustainable performance compared to those without such strategies (Zhou et al., 2019; Solovida & Latan, 2017). ES reflects a firm's strategic commitment to reducing environmental impact while enhancing operational sustainability. As argued by the NRBV, firms that embed environmental concerns into their strategies are better positioned to deploy resources in a way that supports eco-innovation, reducing emissions, waste, and inefficiencies (Hart, 1995).

Previous research suggests that business strategy plays a crucial role in driving environmental protection, business operations, and GI (Kraus et al., 2020; Mulaessa & Lin, 2021). Additionally, evidence from the literature highlights that adopting proactive ES can boost GI, particularly in the context of SMEs (Perez et al., 2025). Indeed, Solovida and Latan (2017) claim that organizations implementing environmental strategies tend to achieve superior environmental outcomes compared to those that do not. ES is key to fostering innovation and driving eco-friendly practices, supporting both sustainable development and green creativity (Aftab et al., 2023; Khan et al., 2021). Therefore, we propose the following:

H3. *ES significantly predicts GI.*

2.5. Mediating role of environmental strategy

ES plays a key mediating role between GKOL and GI, driving sustainable development and improving environmental performance. ES involves practices like energy efficiency, resource conservation, and waste reduction, which promote GI and minimize manufacturing impacts (Mansoor & Hussain, 2024; Tian et al., 2023). Moreover, effective integration of green knowledge and ES fosters GI and organizational sustainability (Aftab et al., 2022; Al Halbusi et al., 2025). From the perspective of DCT, strategy is not static; it evolves through sensing and responding to environmental trends (Teece et al., 1997). GKOL builds organizational capacity for transformation, while ES translates this capacity into action. Thus, ES serves as a bridge through which green leadership can produce innovative outcomes, especially in dynamic and resource-constrained environments like SMEs. In this regard, several studies show that GKOL improves performance, while ES further enhances this relationship by driving green creativity and eco-friendly practices (Miah et al., 2024; Abbas & Khan, 2023). Hence, we suggest the following:

H4: *ES mediates the GKOL-GI link.*

2.6. Moderating role of green artificial intelligence capability

Organizations are increasingly undergoing transformation through the integration of AI technologies, particularly within the realm of knowledge management (KM) (Nakash & Bolisani, 2024). AI significantly enhances the use of both explicit and tacit knowledge by enabling its efficient generation, storage, sharing, and application. This transformation not only drives innovation and productivity but also contributes to a sustainable competitive advantage in today's knowledge-based economy (Abdulmuhsin et al., 2025; Ferreira et al., 2024). AI-powered KM systems are particularly effective at structuring organizational knowledge for optimized access and dissemination, thereby fostering KOL (Bagherimajd & Khajedad, 2024). While KOL has traditionally supported collaboration and sustainability through knowledge sharing (Shamim et al., 2019), conventional approaches often fall short in fast-changing or temporary organizational environments (Thamhain, 2004; Tyssen et al., 2013). AI addresses these limitations by processing vast amounts of data, generating predictive insights, and automating knowledge flows, which enhances KM efficiency while aligning organizational activities with green and sustainable objectives (Chen et al., 2025; Paschen et al., 2020). Moreover, AI plays a vital role in accelerating innovation. It allows for virtual testing of green technologies, minimizes dependency on resource-intensive prototyping, and fosters continuous improvement through iterative design (Wang et al., 2023; Yin & Yu, 2022). Beyond its technical capabilities, AI strengthens leadership decision-making by providing transparent, data-driven insights that cultivate trust, accountability, and cross-functional collaboration (Pellicano et al., 2018).

Emerging research in entrepreneurship has further underscored the value of AI for SMEs. For instance, Shepherd and Majchrzak (2022) demonstrate how AI enhances entrepreneurial cognition and decision-making, offering both new opportunities and threats that necessitate capable and sustainability-focused leadership. Ueasangkomsate (2025) highlights the socio-technical dimensions of digital transformation in Thai food SMEs, which share common challenges with Malaysian manufacturing SMEs, such as resource scarcity and workforce digital readiness. Complementing these insights, Schenkel et al. (2025) propose a

theory of entrepreneurial orientation and technology fit, which supports the idea that achieving GI requires a synchronized alignment between ES and GAIC.

Against this backdrop, GKOL emerges as a vital driver of green innovation in organizations. However, its impact is significantly amplified when paired with GAIC. As an advanced organizational capability, GAIC enables firms to leverage sustainable technologies and data analytics to monitor environmental indicators, predict trends, and automate eco-friendly practices (Wilson et al., 2023). Leaders who are equipped with green knowledge can harness these AI capabilities to advance sustainability goals, foster innovative solutions, and enhance the development of environmentally responsible products (Al Halbusi et al., 2025). This integration empowers strategic decision-making and ensures alignment with organizational sustainability objectives, thereby enhancing the effectiveness of GI initiatives (Abdelfattah et al., 2025).

GAIC thus plays a pivotal moderating role in the relationship between GKOL and GI. As a dynamic capability, GAIC enhances an organization's adaptive capacity by leveraging real-time environmental insights, forecasting environmental trends, and optimizing resource allocation. Based on DCT, GAIC enhances the effectiveness of GKOL, which provides strategic direction and a green knowledge foundation for sustainable transformation, as articulated in the NRBV. By accelerating the implementation of green initiatives, GAIC enables organizations to act quickly based on green insights. Thus, the synergy between GKOL and GAIC leads to more agile, evidence-based, and impactful sustainability strategies. Accordingly, we propose the following hypothesis:

H5. *GAIC moderates the GKOL-GI link.*

Based on the above, Figure 1 shows the conceptual framework developed for this study.

3. Methodology

3.1. Participants and procedure

This study employed a quantitative cross-sectional survey involving 300 respondents, comprising business owners, CEOs, senior managers, and operational managers from SMEs located in the northern region of Malaysia. The region was chosen for its strategic importance in Malaysia's industrial and environmental development agenda, as it hosts a high concentration of SMEs and industrial parks (Al Koliby et al., 2024a). Additionally, manufacturing SMEs in this area encounter significant challenges such as resource limitations, environmental regulations, and the need for digital transformation, making it a suitable context for investigating green leadership and innovation dynamics (Jaish et al., 2023; Musa & Chinniah, 2016). A total of 219 valid responses were received, yielding a valid response rate of 73%, with incomplete or missing responses excluded from analysis. All participants provided informed consent before responding. In Malaysia, SMEs are categorized based on industry type. Manufacturing SMEs are defined as firms with a sales turnover below RM50 million or fewer than 200 full-time employees, while services and other industries classify SMEs as those with a sales turnover under RM20 million and fewer than 75 full-time employees (SME Corporation Malaysia, 2018).

A simple random sampling method was employed, drawing participants from two primary sources: the SME Corporation Malaysia database (2021) and the Federation of Malaysian Manufacturers (FMM) directories (2019, 2020). The response rate fell within an acceptable range; however, the potential for non-response bias was carefully considered. To assess its impact, early and late respondents were compared across key demographic variables, revealing no statistically significant differences. This suggests that non-response bias did not substantially affect the study's findings. To determine the minimum required sample size and

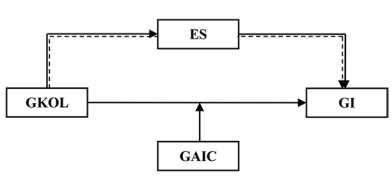

Figure 1. Research model.

assess statistical power, G*Power software (Faul et al., 2007) was used. By setting a 5% significance level, 80% statistical power, and an effect size of 0.15 (Cohen et al., 2013), the minimum required sample size was calculated as 129 cases. Since the study collected 219 valid responses, this sample size was deemed sufficient. Prior research has shown that SMEs typically yield response rates of around 10% (Ho et al., 2016), aligning with the broader expected response range for social science research (5%–35%) (Sekaran & Bougie, 2016). Furthermore, previous studies on Malaysian SMEs have reported average response rates of approximately 20% (Al Koliby et al., 2022, 2024a; Kanapathy & Khan, 2012).

To ensure randomness and mitigate selection bias, a systematic randomization process was employed, where every nth firm from the compiled SME directory was selected. Additionally, to enhance representation and minimize bias associated with directory-based sampling, the sample was stratified by industry sub-sectors, ensuring diverse SME participation. This approach ensured that each SME had an equal chance of selection (Sekaran & Bougie, 2010). Ultimately, the final sample exceeded the recommended threshold of 100 responses (Hair et al., 2019), further confirming its adequacy for statistical analysis. Table 1 lists the demographics characteristics of the participants.

3.2. Questionnaire development and instrument

The self-administered questionnaire encompassed two sections: respondents' demographic profiles and respondents' perception of the relationships between GKOL, ES, GAIC, and GI of the manufacturing SMEs. The Supplementary Appendix shows the measurement items of the instruments employed. GKOL was measured using six-items obtained from Naqshbandi and Jasimuddin (2018). As this scale was originally developed to measure KOL, the six items were modified to assess GKOL.GAIC was measured using six-items adopted from Salehzadeh et al. (2024). ES was measured using three-items adopted from Ko and Liu (2017). Finally, GI was measured using four-items adopted from Al-Swidi et al. (2024). Supplementary Appendix 1 shows the constructs, and the items used to measure them. These items were designed on a five-point Likert scale, i.e. 1= strongly Disagree and 5 = strongly Agree.

3.3. Multicollinearity and common method bias

To assess multicollinearity, we examined the Variance Inflation Factors (VIF) for each predictor variable. All VIF values were well below the commonly accepted threshold of 3.3, indicating that multicollinearity is not a concern in this analysis (Friedman & Wall, 2005).

Table 1. Participant demographics characteristics.

Description	Frequency
Position	
CEO	53
Marketing executive	46
Operation executive	27
R&D executive	48
Quality manager	45
No. of employees	
Between 5 and 74	105
Between 75 and 200	114
Established (years)	
5–10	70
11–15	94
>15	55
Sub-Sector Industry	
Food and beverage	73
Paper and paper products	11
Non-metallic mineral products	3
Products and machinery	14
Automobile and motoring products	5
Textile and wearing apparel	15
Wood and wood products	46
Electrical and electronic	22
Iron and steel	30

Source: Derived from the statistical analysis of this study (SPSS 25).

Additionally, to reduce potential common method bias (CMB) arising from the self-reported nature of the data, several procedures were implemented during the data collection process, where respondent anonymity and confidentiality were strictly maintained to minimize social desirability bias. In addition, all measurement items were adopted from well-established and previously validated scales to enhance reliability and conceptual clarity. Moreover, we performed Harman's single-factor test, where the results showed that a single factor did not account for the majority of the variance, suggesting that CMB is unlikely to affect our data (Podsakoff & Organ, 1986). These diagnostic tests confirm the reliability of our dataset for subsequent analyses.

3.4. Statistical technique

In this study, we employed Partial Least Squares Structural Equation Modeling (PLS-SEM) using SmartPLS 4.0 to analyze data from 219 cases. PLS-SEM was chosen for its suitability in handling complex models, small sample sizes, and non-normal data distributions (Henseler & Schuberth, 2023; Sarstedt et al., 2022). This approach aligns with our study's requirements, effectively examining relationships among multiple constructs with the available sample size.

3.5. Ethical statement

"This study received ethical approval from the Ethics Committee of School of International Studies, Universiti Utara Malaysia, Kedah, Malaysia (Approval number: UUM/COLGIS/SoIS/2024/EC101), which guarantees that ethical guidelines are followed for research involving human subjects. The committee granted approval following a thorough review of the research suggestion and questionnaire items."

3.6. Informed consent

Participants were given written informed consent forms that outlined the study's nature and purpose, as well as their rights and responsibilities. Since data was collected through survey, this served as written authorization. Participants were assured that they could withdraw from the study at any time without any repercussions and that their participation in it was entirely voluntary.

4. Results

4.1. Measurement model analysis

The measurement model was tested for reliability and validity, including convergent and discriminant validity. Indicators with outer loadings above 0.70, composite reliability (CR) exceeding 0.70, and average variance extracted (AVE) above 0.50 confirmed internal consistency and convergent validity (Hair et al., 2019).

Discriminant validity was assessed using the HTMT criterion, which requires coefficients to be below 0.85 (Henseler & Schuberth, 2023). All results satisfied these criteria, confirming the reliability and validity of the measurement model, as shown in Tables 2 and 3).

4.2. Structural model analysis

The structural model was analyzed to examine links between constructs. Using bootstrapping with 5,000 resamples, model fit was evaluated via R^2 values and the significance of path coefficients (Hair et al., 2019). The analysis showed that the model explained 42.5% of the variation in ES, 39.5% in GI (see Table 4).

Table 5 highlights the results: GKOL had a significant positive impact on SE ($\beta = 0.330$, $t = 3.840$, $p < 0.05$), supporting H1. Additionally, GKOL significantly influenced GI ($\beta = 0.209$, $t = 2.695$, $p < 0.05$), validating H2. ES also showed a positive effect on GI ($\beta = 0.835$, $t = 13.935$, $p < 0.05$), supporting H3.

4.4. Mediation analysis

Using the bootstrapping approach, the mediation analysis revealed that ES mediates the GKOL–GI relationship ($\beta = 0.274$, $t = 3.782$), as presented in Table 6. These results confirmed that ES partially mediates between GKOL and GI, supporting hypotheses H4.

Table 2. Construct reliability and validity.

Construct	Code	Loading	CR	AVE
GKOL			0.901	0.696
	GKOL1	0.795		
	GKOL2	0.895		
	GKOL3	0.874		
	GKOL4	0.765		
	GKOL5	0.752		
	GKOL6	0.843		
ES			0.920	0.794
	ES1	0.855		
	ES2	0.909		
	ES3	0.907		
GAIC			0.910	0.718
	GAIC1	0.784		
	GAIC2	0.852		
	GAIC3	0.877		
	GAIC4	0.874		
	GAIC5	0.849		
	GAIC6	0.863		
GI			0.923	0.858
	GI1	0.938		
	GI2	0.914		
	GI3	0.787		
	GI4	0.861		

Note: AVE = average variance extracted; CR = composite reliability; GKOL = Green knowledge-oriented leadership; ES = Environmental Strategy; GAIC = Green artificial intelligence capability; GI = Green innovation.
Source: Derived from the statistical analysis of this study (SmartPLS).

Table 3. Discriminant validity by HTMT.

Construct	GKOL	ES	GAIC	GI
GKOL				
ES	0.809			
GAIC	0.686	0.628		
GI	0.849	0.889	0.553	

Source: Derived from the statistical analysis of this study (SmartPLS).

Table 4. Model evaluation indicators.

Construct	R^2	Q^2
ES	0.425	0.458
GI	0.395	0.371

Source: Derived from the statistical analysis of this study (SmartPLS).

Table 5. Direct and interactive effects.

Path	b	SD	T value	*p* value
GKOL→ES	0.330	0.086	3.840	0.000
GKOL→GI	0.209	0.078	2.695	0.000
ES→GI	0.835	0.060	13.935	0.000
GAIC*GKOL→GI	0.080	0.039	3.037	0.018

Source: Derived from the statistical analysis of this study (SmartPLS).

Table 6. Indirect effects.

Path	b	SD	T value	*p* value
GKOL→ES→GI	0.274	0.075	3.782	0.000

Source: Derived from the statistical analysis of this study (SmartPLS).

4.4. Moderation analysis

To identify whether GAIC moderates the GKOL-GI link, the moderation analysis was conducted using the orthogonalization approach. The hypothesis was supported by the absence of zero in the lower and upper confidence intervals and statistically significant *p*value ($p < 0.05$). The results, presented in Table 5, indicate that GAIC significantly moderates the GKOL–GI relationship ($\beta = 0.080$, $t = 3.037$, $p < 0.05$), thereby supporting Hypothesis H5. The interaction graph (Figure 2) further illustrates this result.

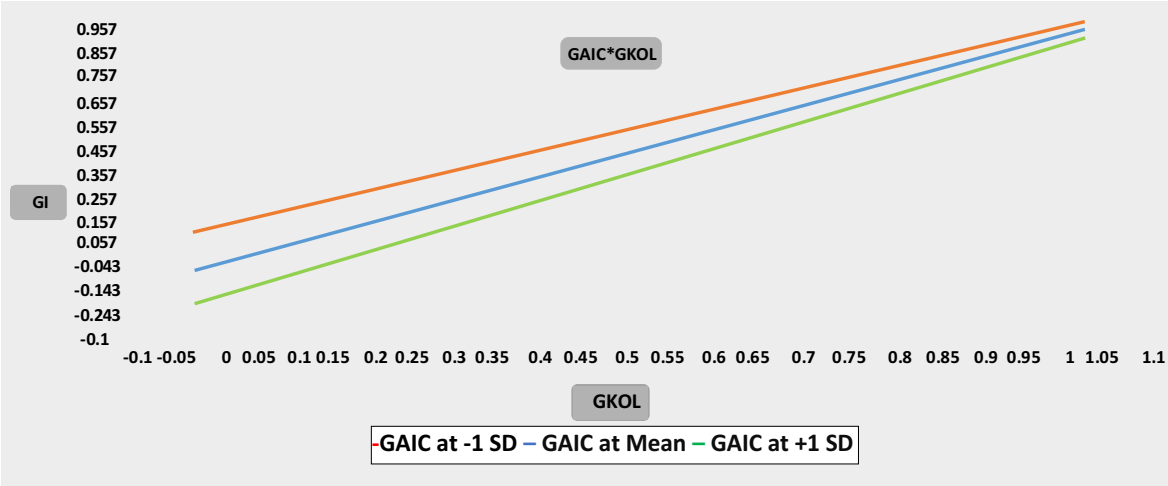

Figure 2. Moderating role of GAIC.

5. Discussion

Guided by the NRBV and DCV theories, this study explores whether GKOL directly affects GI and indirectly affects it via ES, as well as the moderating effect of GAIC. Using survey data from SMEs in Malaysia, the research provides empirical evidence to evaluate the proposed model and its underlying assumptions. Overall, the findings enhance theoretical understanding by demonstrating how ES mediates the relationship between GKOL and GI, while GAIC moderates this link, offering new insights into sustainability-driven leadership and innovation.

Based on the analysis results, GKOL has a significant positive effect on GI. This aligns with earlier studies emphasizing the importance of sustainability-focused leadership in fostering green creativity and innovation (Donate & de Pablo, 2015; Khan et al., 2023) and supports previous findings (e.g. Naqshbandi & Jasimuddin, 2018) that recognize KOL as a key driver of innovation. This connection is grounded in the NRBV, which considers green knowledge a strategic asset that facilitates sustainable innovation, where leaders who promote environmental learning and green knowledge sharing help create an organizational culture that supports innovation. Indeed, KOL provides organizations with essential tools such as employee involvement, collaboration, feedback systems, and effective communication that naturally support innovative and sustainable behaviors (Green et al., 2019). From the perspective of DCT, GKOL enables companies to identify and capitalize on environmental opportunities, motivating employees to pursue eco-friendly solutions through continuous learning and experimentation (Shamim et al., 2019).

The findings also showed that GKOL significantly influences ES. This is in line with studies emphasizing the critical role of leadership in formulating proactive environmental strategies (Al-Faouri, 2023; Hussain et al., 2022). Leaders who prioritize environmental knowledge embed sustainability into organizational strategies and decision-making processes. This supports the NRBV proposition that leadership drives the mobilization of strategic resources such as policies, goals, and practices toward sustainable outcomes (Hart, 1995) and also confirms DCT, where GKOL helps firms reconfigure their internal processes and strategies to respond to environmental demands.

Additionally, the results also revealed that ES significantly predicts GI. This aligns with prior studies that associate proactive environmental strategies with increased levels of green product and process innovation (Kraus et al., 2020; Mulaessa & Lin, 2021). It also reinforces the findings of Perez et al. (2025), who highlighted a strong positive link between ES and GI, particularly in the context of SMEs. Similarly, it supports Solovida and Latan (2017) conclusion that organizations implementing environmental strategies tend to achieve superior environmental outcomes compared to those that do not. This evidence suggests that embedding sustainability into strategic planning enhances a firm's ability to pursue eco-innovative initiatives. In line with the NRBV, such strategies facilitate the efficient allocation of environmental resources, contributing to a lasting competitive advantage (Kraus et al., 2020).

Importantly, ES was found to partially mediate the relationship between GKOL and GI. This indicates that while GKOL directly influences GI, its impact is significantly enhanced when channeled through a formalized environmental strategy. ES acts as the mechanism that aligns leadership intentions with innovation practices. From the NRBV perspective, this result supports the view that resource deployment such as leadership influence must be structured through strategy to generate performance outcomes. This also aligns with DCT, as ES enables firms to align and reconfigure resources in response to external sustainability pressures, facilitating more effective innovation (Rehman et al., 2021; Abbas & Khan, 2023).

Finally, the moderation analysis revealed that GAIC significantly moderates the relationship between GKOL and GI. This finding suggests that leadership's impact on innovation is amplified when firms possess advanced AI capabilities that support environmental monitoring, predictive analytics, and automated green solutions. This finding also aligns with emerging literature on the role of AI in driving innovation and sustainability (Abdulmuhsin et al., 2025; Salehzadeh et al., 2024). This supports the DCT proposition that GAIC is considered a dynamic technological capability that enhances a firm's adaptability. It allows organizations to respond quickly to sustainability challenges by enabling data-driven decision-making and resource optimization. While the study highlights GAIC as a critical moderating factor that enhances the relationship between GKOL and GI, it is important to acknowledge that GAIC was treated as a broad construct encompassing AI-enabled technologies for sustainability decision-making and innovation. However, the technological depth, such as the specific types of AI systems deployed (e.g. machine learning, predictive analytics, or automation tools), and the practical challenges SMEs face in implementing such technologies were not explored in detail, which may affect the generalizability of the findings, as SMEs are likely to be at different levels of digital readiness and technological capability.

6. Conclusion

This empirical investigation was conducted on manufacturing SMEs in Malaysia, an emerging economy facing pressing environmental and technological challenges. The results of this study enrich our understanding of the subject and address a notable research gap concerning the mechanisms through which leadership drives green outcomes in resource-constrained contexts. This study introduces a new model that offers a broader perspective on the GKOL–GI relationship by examining the mediating role of ES and the moderating role of GAIC. The findings are consistent with the NRBV and DCT, as they provide empirical support for the positive and significant impact of GKOL on GI through the mediating effect of ES, particularly when GAIC is present. Specifically, the findings demonstrate that GKOL significantly enhances ES, which in turn positively influences GI. Furthermore, ES serves as a partial mediator between GKOL and GI, highlighting the strategic pathway through which leadership fosters innovation. Notably, GAIC strengthens the direct effect of GKOL on GI, emphasizing the enabling role of AI-driven capabilities in advancing GI among Malaysian SMEs.

6.1. Theoretical implications

This study offers several important theoretical contributions by advancing an integrated understanding of how leadership, strategy, and digital capabilities interact to foster GI in SMEs. First, it extends the NRBV by positioning GKOL as a unique, valuable, and inimitable internal capability that enables firms to generate and leverage environmental knowledge for strategic advantage. By doing so, it emphasizes the strategic role of leadership in mobilizing intangible green knowledge assets that are often overlooked in traditional resource-based frameworks. Second, the study deepens the application of DCT by demonstrating that GAIC functions as a dynamic capability that enhances a firm's responsiveness to environmental challenges. Specifically, GAIC enables firms to sense, seize, and reconfigure their environmental innovation processes through data-driven insights and AI-enabled decision-making. This contribution is particularly significant as it highlights how SMEs can transcend their resource limitations by leveraging digital technologies, offering a nuanced extension of DCT in sustainability-oriented contexts. Third, the study makes a significant contribution by integrating NRBV and DCT into a unified framework to explain the mechanisms through which GKOL translates into GI through the mediating role of ES and the moderating role

of GAIC. Previous literature has examined these elements in isolation; however, this study bridges these theoretical domains, revealing that sustainability-oriented innovation is not solely dependent on leadership but requires strategic alignment and technological support due to organizational and technological constraints (Ahmad et al., 2024). By showing that ES partially mediates and GAIC moderates the GKOL–GI relationship, the study contributes to bridging existing conceptual gaps. It offers a more comprehensive understanding of how green knowledge can be effectively translated into innovation outcomes through strategic alignment and technological support. This theoretical integration, hence, helps clarify mixed findings in prior research regarding the impact of leadership on innovation outcomes in resource-constrained environments and offers a new perspective on the leadership–strategy–technology nexus in sustainability research, thereby strengthening the theoretical grounding of both NRBV and DCT (Aftab et al., 2022; Wang et al., 2022). Finally, the study contributes to the leadership and innovation literature in developing country contexts by providing empirical evidence from Malaysian manufacturing SMEs. This contextual specificity adds theoretical richness, as it shows that the interplay of leadership, strategy, and digital capability is shaped by institutional and technological challenges prevalent in emerging economies. Therefore, the study lays the groundwork for future comparative research that can explore the generalizability of these relationships across different cultural and institutional environments.

6.2. Practical implications

This study presents several practical recommendations for SME managers, policymakers, and sustainability advocates especially in developing economies like Malaysia, where limited resources and regulatory demands present hurdles to sustainable transformation. First, the findings underscore the critical role of GKOL in driving GI. SME leaders should cultivate a culture that promotes environmental awareness, knowledge sharing, and employee empowerment to support green practices. Leadership development programs should incorporate sustainability-oriented training to equip managers with the vision and capabilities needed to champion green initiatives. These efforts can align organizational values with sustainability goals and enhance long-term competitiveness. Second, since ES acts as a key mechanism linking leadership to innovation, SMEs should move beyond informal or reactive green initiatives and instead embed structured environmental strategies into their core business plans. This involves setting clear sustainability objectives, developing green performance metrics, and ensuring compliance with environmental regulations. Strategic planning with an environmental focus will help SMEs systematically channel leadership intentions into actionable green innovations. Third, the moderating role of GAIC indicates that digital technologies can enhance the effectiveness of green leadership in promoting innovation. Therefore, SMEs should invest in AI tools—such as predictive analytics, machine learning, and environmental monitoring systems that support green decision-making and optimize resource usage. However, SMEs must balance technological advancement with sustainability by focusing on energy efficient AI systems that align with environmental objectives. This not only supports innovation but also ensures that the environmental benefits of digital transformation outweigh potential ecological costs. Furthermore, the findings underscore the importance of an integrated approach that combines leadership, strategic planning, and technological infrastructure to drive sustainability. Hence, practitioners should focus on building internal capacities while also collaborating with external stakeholders such as research institutions, government bodies, and green technology providers to co-create sustainable solutions tailored to industry needs. Finally, policymakers and industry associations in Malaysia should play a proactive role in enabling SME green transformation by offering incentives for green leadership initiatives, subsidizing AI adoption, and providing technical training programs. Public-private partnerships can bridge the resource and knowledge gaps faced by SMEs, facilitating access to advanced technologies and sustainability expertise.

6.3. Limitations and future research

While the study provides significant contributions, it is not without limitations, which present opportunities for future exploration. First, the study is geographically and sectorally limited to manufacturing SMEs in Northern Malaysia. Although this context offers valuable insights into how resource-constrained

firms can leverage leadership and digital capabilities for GI, the findings may not be fully generalizable to other sectors or regions. Institutional pressures, technological readiness, and cultural variations across countries or industries may influence the relationships observed. Future studies should consider expanding the sampling frame to include SMEs from different regions and sectors—such as services, agriculture, or energy—within Malaysia and beyond to enhance the external validity of the model. Second, the research employed a cross-sectional design, capturing data at a single point in time. While this design offers a snapshot of the relationships between GKOL, ES, GAIC, and GI, it does not allow for inferences about causality or long-term effects. To build on this study, future research could adopt a longitudinal approach to track the evolution of these relationships over time, thereby offering deeper insights into the sustained impact of green leadership and technological capability on innovation outcomes. Additionally, integrating mixed-method designs—including qualitative interviews or case studies—could further enrich our understanding of the underlying mechanisms and contextual nuances. Third, the study relied on self-reported data from a single respondent per firm, which may introduce CMB despite the precautions taken. Future research could enhance validity by collecting multi-source data (e.g. including environmental performance reports or third-party assessments) or by incorporating objective innovation performance metrics. Fourth, while this study explored the moderating role of GAIC, it did not delve into specific AI technologies being adopted or assess the varying maturity levels of these technologies within firms. Future studies could investigate which types of AI technologies (e.g. machine learning, predictive analytics, or environmental monitoring tools) are most effective in enhancing sustainability outcomes and the challenges of their adoption. Fifth, while this study employed the NRBV and DCT frameworks to explore the internal drivers of GI, integrating other theoretical frameworks, such as stakeholder theory and institutional theory, could enhance the depth and breadth of understanding regarding the proposed relationships. Future research may benefit from incorporating these complementary perspectives. Finally, the model could be extended by exploring additional mediating and moderating variables such as organizational culture, employee green behavior, or external stakeholder engagement—that may influence the leadership innovation nexus. Exploring these dynamics could provide a more holistic view of how SMEs transition toward sustainability-driven innovation in increasingly complex environments.

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Author contribution statement

CRediT: **Ibraheem Saleh AlKoliby**: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing; **Abdullah Kaid Al-Swidi**: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing; **Mohammed A. Al-Hakimi**: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing; **Shaima Ali Galeb Farhan**: Conceptualization, Methodology, Resources, Software, Writing – review & editing.

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About the authors

Ibraheem Saleh Al Koliby is an Assistant Professor of Marketing at the Department of Marketing & Production, Thamar University, Yemen. His research explores themes such as entrepreneurship, innovation, knowledge management, and sustainability.

Abdullah Kaid Al-Swidi is an Associate Professor of Management in the Department of Management and Marketing, College of Business and Economics, Qatar University. His research work revolves around business excellence, green operations, and green behavior.

Mohammed A. Al-Hakimi is an Assistant Professor of production and operations management at the Department of Marketing & Production, Thamar University, Yemen. His research interests span sustainable operations, supply chain management, digital technologies, and business strategy.

Shaima Ali Galeb Farhan is a PhD candidate at the School of Management, University of Science and Technology of China, Hefei, Anhui, China. Her research interests span business strategy, operations research, and business analytics.

ORCID

Mohammed A. Al-Hakimi http://orcid.org/0000-0001-9923-1439

Data availability statement

The data used in this study are available on request from the corresponding author.

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